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HISTORY · National Movement I: founding of the Congress; Moderates and Extremists

UNDERSTAND IT FIRST

What is it? By the 1880s, educated Indians across the country began to feel that they were **one nation** with **common grievances** against British rule. They formed **associations**, then an **all-India body**, the **Indian National Congress (1885)**. In its first twenty years the Congress was led by **Moderates**, who trusted British justice and asked for **reforms through petitions**. After **1905**, a younger group, the **Extremists (Assertive nationalists)**, demanded **swaraj** and used **mass action**: boycott and swadeshi. Learn it as **Why nationalism rose → early associations → founding of the Congress → Moderates → rise of Extremists → Surat split**.

Early national movement

1. Why Indian nationalism rose

Area	Causes
Politics	**Political unification** under one administration; **Lytton's** measures: the **Delhi Durbar (1877)** held during famine, **Vernacular Press Act and Arms Act (1878)**, lowering the **ICS age limit to 19**; the **Ilbert Bill controversy (1883)**, where Europeans opposed trial by Indian judges, showing **racial discrimination**.
Economy	**Drain of wealth**, ruin of crafts, heavy land revenue and **famines** made **economic exploitation** clear; the **railways, telegraph and post** linked people across India.
Society	**Western education** spread ideas of **liberty and democracy**, and English became a **common language**; the **press** (The Hindu 1878, Amrita Bazar Patrika, Kesari) carried these ideas.
Religion and culture	**Reform movements** and the **rediscovery of India's past** (by Indian and European scholars) built **pride and self-confidence**.

2. Associations before the Congress

Association	Year, place, founders
Landholders' Society	**1838**, Calcutta; **Dwarkanath Tagore**; first political body, for zamindars' interests.
British Indian Association	**1851**, Calcutta; petitioned Parliament on the 1853 Charter.
Madras Native Association and **Bombay Association**	**1852**.
East India Association	**1866**, **London**; **Dadabhai Naoroji**; won British opinion for India.

Poona Sarvajanik Sabha	**1870**; associated with **M.G. Ranade**.
Indian Association	**1876**, Calcutta; **Surendranath Banerjee** and **Ananda Mohan Bose**; campaigned on the ICS age; held the **Indian National Conference (1883, 1885)**, which merged into the Congress.
Madras Mahajana Sabha and **Bombay Presidency Association**	**1884** (M. Viraraghavachari, G. Subramania Iyer, **P. Ananda Charlu**) and **1885** (Pherozeshah Mehta, K.T. Telang, Badruddin Tyabji).

3. Founding of the Indian National Congress

- **First session: 28–31 December 1885** at **Gokuldas Tejpal Sanskrit College, Bombay** (planned for **Poona**, shifted because of **cholera**); **72 delegates**; president **W.C. Bonnerjee**; organised by **A.O. Hume**, a retired British civil servant; Viceroy **Lord Dufferin**; the name 'Congress' was suggested by **Dadabhai Naoroji**.
- **'Safety valve' theory:** Hume founded the Congress so that Indian **discontent could escape safely** and not become another 1857 (stated by **Lala Lajpat Rai** in 'Young India', 1916). Historians like **Bipan Chandra** say Indian leaders used Hume as a **'lightning conductor'**: an Englishman at the head would **avoid official suspicion**.

Session	President and note
1885 Bombay	**W.C. Bonnerjee**
1886 Calcutta	**Dadabhai Naoroji** (president three times: 1886, 1893, 1906)
1887 Madras	**Badruddin Tyabji**, first **Muslim** president
1888 Allahabad	**George Yule**, first **English** president
1891 Nagpur	**P. Ananda Charlu**, a **Telugu** leader of the Madras Presidency
1905 Benares	**G.K. Gokhale**; supported swadeshi against the Partition
1906 Calcutta	**Naoroji**; **swaraj** declared the goal; resolutions on **swadeshi, boycott, national education**
1907 Surat	**Rash Behari Ghosh**; the **split**

4. Moderates and Extremists compared

Point	Moderates (1885–1905)	Extremists (1905–1920)
Leaders	Naoroji, Pherozeshah Mehta, **Gokhale**, Surendranath Banerjee, Ranade, W.C. Bonnerjee, Badruddin Tyabji	**Lal-Bal-Pal**: Lala Lajpat Rai, **Bal Gangadhar Tilak**, Bipin Chandra Pal; **Aurobindo Ghosh**
Goal	**Reforms** and more Indians in councils and services, within British rule	**Swaraj** (self-rule); Tilak: **'Swaraj is my birthright and I shall have it'**
Methods	Constitutional: **petitions, prayers, protests** (speeches, memoranda); faith in **British justice**	**Boycott, swadeshi, national education, passive resistance**; self-

		reliance
Base	Educated **urban middle class**	Wider: **students, youth, lower middle class**, use of **festivals** (Tilak's **Ganapati 1893**, **Shivaji 1895**)
Achievements	**Economic critique (drain)**; **Indian Councils Act 1892**; political education; Naoroji elected to the **British Parliament (1892)**; Gokhale's **Servants of India Society (1905)**	Made the movement **mass-based**; **swadeshi industry**; **national schools**; **pride and courage**
Limits	Did not reach the **masses**; British gave little	Some use of **religious symbols** limited appeal to all communities; no clear **plan for mass action**

5. Why extremism grew, and the Surat split

- **Causes: failure of Moderate methods**; **Curzon's** policies (**Calcutta Corporation Act 1899**, **Universities Act 1904**, **Partition of Bengal 1905**); the **plague and famine of 1896–97** and harsh plague measures (the **Chapekar brothers** shot the plague officer **Rand**, 1897; Tilak jailed for sedition); **world events**: Ethiopia beat **Italy (1896)**, Japan beat **Russia (1905)**; influence of **Vivekananda, Dayananda** and **Aurobindo's 'New Lamps for Old' (1893)**.
- **Surat split (1907):** quarrel over the **president** (Moderates chose **Rash Behari Ghosh**; Extremists wanted **Lajpat Rai or Tilak**) and fear that Moderates would **drop the 1906 resolutions** on swadeshi and boycott. The Congress split; the British then **crushed the Extremists** (Tilak deported to **Mandalay, 1908–14**, where he wrote **Gita Rahasya**) and **wooed the Moderates** with the **1909 reforms**. The two groups **reunited at Lucknow in 1916** (Day 20).
- **Telugu link: P. Ananda Charlu** presided over the Congress (**1891**); **Nyapathi Subba Rao** of Rajahmundry was one of the founders of **The Hindu (1878)**; the Telugu districts sent delegates from the first sessions, and Bipin Chandra Pal's **1907 tour of Andhra** (Day 20) carried the swadeshi message.

EXAM TRAP

Associations: Landholders' 1838, British Indian 1851, East India Assoc. 1866 (London), Poona Sarvajanik 1870, Indian Association 1876, Madras Mahajana 1884, Bombay Presidency 1885; **INC: 28 Dec 1885, Bombay, 72 delegates, W.C. Bonnerjee, Hume, Dufferin**; **safety valve** (Lajpat Rai); **1886** Naoroji, **1887** Tyabji, **1888** Yule, **1891** Ananda Charlu, **1906** swaraj goal, **1907** Surat split; **Moderates**: 3 Ps, 1892 Act; **Extremists**: Lal-Bal-Pal, Aurobindo, Kesari, Ganapati 1893, Shivaji 1895; Curzon, Partition **1905**; Tilak to Mandalay **1908**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the first session of the Indian National Congress:
1. It was held in Bombay in December 1885.

2. It was attended by 72 delegates.

3. It was presided over by W.C. Bonnerjee.

4. It was originally planned to be held in Calcutta.

Which of the above are correct?

A. 1 and 2 only

B. 2, 3 and 4 only

C. 1, 2, 3 and 4

D. 1, 2 and 3 only

Answer: D. It was planned for Poona and shifted to Bombay because of cholera.

PRACTICE

Match List-I (Congress session) with List-II (president):

List-I: A. 1885 Bombay B. 1886 Calcutta C. 1887 Madras D. 1907 Surat

List-II: 1. Rash Behari Ghosh 2. Badruddin Tyabji 3. Dadabhai Naoroji 4. W.C. Bonnerjee

A. A-3, B-4, C-2, D-1

B. A-4, B-2, C-3, D-1

C. A-4, B-3, C-2, D-1

D. A-4, B-3, C-1, D-2

Answer: C. Bonnerjee 1885; Naoroji 1886; Tyabji 1887; Rash Behari Ghosh 1907.

PRACTICE

Which one of the following statements about the Moderates is NOT correct?

A. They believed in the British sense of justice and fair play.

B. They used boycott of British goods and passive resistance as their main methods.

C. They demanded that the ICS examination be held in India as well as in England.

D. The Indian Councils Act of 1892 partly met their demands.

Answer: B. Boycott and passive resistance were Extremist methods; the Moderates used petitions, prayers and protests.

PRACTICE

Assertion (A): Extremism grew within the Congress in the early years of the 20th century.
Reason (R): Curzon's policies, especially the Partition of Bengal, and the failure of Moderate methods angered younger nationalists.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. R gives the main reasons for the rise of extremism.

PRACTICE

A view holds that the Congress was founded so that Indian discontent could be released safely, preventing another violent revolt like 1857. What is this view called?

A. The drain theory

B. The safety-valve theory

C. The downward filtration theory

D. The two-nation theory

Answer: B. The safety-valve theory, voiced by Lala Lajpat Rai (1916).

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the causes of nationalism and the seven early associations. 2. Write the founding of the Congress and the session table. 3. Write the Moderates vs Extremists table, causes of extremism and the Surat split.

POLITY · Lokpal and Lokayukta

UNDERSTAND IT FIRST

What is it? A **Lokpal** ('protector of the people') is an **ombudsman**: an independent authority that **hears complaints of corruption** against **public servants**, including the **highest ones**. Ordinary police and courts are slow and may face **political pressure** when a minister or senior officer is accused; an ombudsman

gives citizens an **independent** forum. The idea came from **Sweden (1809)**. India's **Lokpal** works at the **Centre**; each State has a **Lokayukta**. Learn it as **origin → composition and selection → jurisdiction → powers → Lokayukta → criticism**.

Lokpal at a glance

1. Origin

- **Ombudsman:** first set up in **Sweden (1809)**; spread to Finland, Denmark, Norway, New Zealand and the UK. In India, **L.M. Singhvi** coined the words **'Lokpal' and 'Lokayukta' (1963)**; the **First Administrative Reforms Commission (1966)**, chaired by **Morarji Desai**, recommended them.
- **Long wait:** Lokpal Bills were introduced from **1968** onwards but **lapsed many times**; the **2011 anti-corruption movement** led by **Anna Hazare** pushed Parliament to pass the **Lokpal and Lokayuktas Act, 2013** (in force **16 January 2014**). First Lokpal: **Justice Pinaki Chandra Ghose (March 2019)**; **Justice A.M. Khanwilkar** became chairperson in **2024**.

2. Composition, selection and removal

Point	Details
Members	**Chairperson** (a former **CJI** or **Supreme Court judge**, or an eminent person with 25 years' experience in anti-corruption, public administration, law etc.) + **up to 8 members**; **half judicial**; at least **half** from **SC, ST, OBC, minorities and women**.
Selection committee	**Prime Minister** (chair), **Lok Sabha Speaker**, **Leader of the Opposition** in the Lok Sabha (or leader of the largest opposition party), **CJI or a Supreme Court judge** he nominates, and an **eminent jurist** nominated by the President on the others' advice; a **search committee** prepares the panel. Appointed by the **President**.
Term and removal	**5 years or age 70**; not eligible for re-appointment or most later government posts. Removed by the **President** for misbehaviour after a **Supreme Court inquiry**, on a petition signed by at least **100 MPs** (or on a reference by the President).

3. Jurisdiction and powers

- **Covers:** the **Prime Minister**, **Union ministers**, **MPs**, **Group A, B, C and D** central officers, and bodies or **NGOs** funded by government or receiving foreign donations above **₹10 lakh**. **Prime Minister: limits**: no inquiry on **international relations, external and internal security, public order, atomic energy and space**; an inquiry needs the **full bench** and **two-thirds** of members, held **in camera**. **MPs** are not covered for what they **say or vote in Parliament** (Art 105).
- **Not covered: judges** of the Supreme Court and High Courts. **Powers:** has an **inquiry wing** and a **prosecution wing**; **powers of a civil court**; can order **attachment of property** gained by corruption; **superintendence over the CBI** for cases it refers; trials by **special courts**. Complaints must be made within **7 years** of the offence; **false complaints** can be punished.

4. Lokayukta in the States

- **Section 63** of the 2013 Act requires **every State** to set up a **Lokayukta by its own law within one year**; the **structure, powers and coverage** are left to each State, so they **differ** widely. **Maharashtra** set up the **first Lokayukta (1971)**. **Andhra Pradesh**: the **AP Lokayukta and Upa-Lokayukta Act, 1983**; the Lokayukta is appointed by the **Governor** after consulting the **Chief Justice of the High Court** and the **Leader of the Opposition**, and covers **State ministers and public servants**.

5. Criticism and evaluation

- **Strengths:** independent body with judicial members and **social diversity**; covers even the **PM**; **time-bound** process; **property attachment**. **Weaknesses:** only **statutory** (not constitutional); the **selection committee** can be dominated by the government side; the **judiciary** is excluded; **delays** in appointments and few convictions; **complainants** may fear exposure; State **Lokayuktas vary** in strength.

EXAM TRAP

Sweden 1809; **Singhvi 1963**; **ARC 1966 (Morarji Desai)**; **Anna Hazare 2011**; **Act 2013, in force 16 Jan 2014**; **chair + 8**, half judicial, half SC/ST/OBC/minorities/women; committee: **PM, Speaker, LoP, CJI or judge, eminent jurist**; **5 yrs/70**; removal: **President after SC inquiry, 100 MPs**; PM covered with **limits**; judges **excluded**; NGOs > **₹10 lakh** foreign funds; complaint within **7 years**; first Lokpal **P.C. Ghose 2019**; **Khanwilkar 2024**; **Sec 63** Lokayukta; **Maharashtra 1971**; **AP Act 1983**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements about the Lokpal:
1. The First Administrative Reforms Commission (1966) recommended the setting up of the Lokpal and Lokayuktas.
2. It was established under the Lokpal and Lokayuktas Act, 2013.
3. It has jurisdiction over the judges of the Supreme Court.
4. Justice Pinaki Chandra Ghose was the first Lokpal.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 4 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. Judges of the Supreme Court and High Courts are outside the Lokpal's jurisdiction.

PRACTICE

Match List-I with List-II:

List-I: A. Origin of the ombudsman B. Coined the term 'Lokpal' C. First State to set up a Lokayukta D. Chairman of the First ARC

List-II: 1. Morarji Desai 2. Maharashtra 3. L.M. Singhvi 4. Sweden

A. A-3, B-4, C-2, D-1

B. A-4, B-2, C-3, D-1

C. A-4, B-3, C-1, D-2

D. A-4, B-3, C-2, D-1

Answer: D. Sweden; Singhvi; Maharashtra; Morarji Desai.

PRACTICE

Which one of the following statements about the Lokpal is NOT correct?

A. It can inquire into allegations against the Prime Minister relating to national security without any restriction.

B. At least half of its members must be from SC, ST, OBC, minorities or women.

C. Its chairperson can be removed by the President after an inquiry by the Supreme Court.

D. Members hold office for five years or until the age of 70.

Answer: A. Matters of external and internal security, public order, international relations, atomic energy and space are excluded for the PM.

PRACTICE

Assertion (A): Under the 2013 Act, every State had to set up a Lokayukta by its own law within one year.

Reason (R): The 2013 Act lays down a uniform structure and powers for the Lokayuktas of all States.

Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: C. A is true (Sec 63). R is false: each State decides the structure and powers, so they differ.

PRACTICE

A citizen has evidence that a Union minister took a bribe while awarding a central contract. Which body should he approach?

A. The State Lokayukta

B. The Lokpal

C. The National Human Rights Commission

D. The Comptroller and Auditor General

Answer: B. Union ministers are within the Lokpal's jurisdiction.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the origin: Sweden, Singhvi, ARC, Anna Hazare, the 2013 Act, first Lokpal. 2. Write composition, selection committee, term and removal. 3. Write jurisdiction with the PM's limits, powers, Lokayukta, and criticism.

SCIENCE · Space technology: launch vehicles, satellites and ISRO missions

UNDERSTAND IT FIRST

What is it? A **satellite** is a machine that goes **round the Earth (or another body) in an orbit**. To put it there, a **rocket (launch vehicle)** must give it a very high speed (about **8 km per second** for a low orbit). Satellites help **daily life**: **TV and phone links**, **weather forecasts and cyclone warnings**, **maps, crops and disasters** (remote sensing), and **navigation**. India's space work began **to serve development**, as **Vikram Sarabhai** said. Learn it as **history → launch vehicles → orbits → satellite families → missions → centres and private players**.

1. History

- **INCOSPAR (1962)** under **Vikram Sarabhai** (father of the Indian space programme); first rocket (**Nike-Apache**) from **Thumba (TERLS), Kerala, on 21 November 1963**; **ISRO** formed on **15 August 1969**; **Department of Space and Space Commission (1972)**.
- **Aryabhata**, India's first satellite, launched by a **Soviet** rocket on **19 April 1975**; **Bhaskara (1979)**; **SLV-3** put **Rohini** in orbit in **1980** (project director **A.P.J. Abdul Kalam**), making India the **sixth** nation with its own launch capability.

2. Launch vehicles and orbits

Indian launch vehicles

Orbit	Height and use
Low Earth orbit (LEO)	About **200–2,000 km**; the space station, crewed flights, some Earth observation.
Sun-synchronous polar orbit (SSPO)	About **600–800 km**, passing over the poles; crosses each place at the **same local time** every day, so images have the **same lighting**; **remote sensing** (IRS, Cartosat); usually launched by **PSLV**.
Geostationary orbit (GEO)	**35,786 km** above the **equator**; goes round **once in 24 hours** with the Earth, so it seems **fixed over one place**; **communication, TV, weather** (INSAT, GSAT); reached via a **geosynchronous transfer orbit (GTO)** by **GSLV or LVM3**.
Lagrange point L1	About **1.5 million km** from Earth towards the Sun; a spacecraft there can **watch the Sun without eclipses** (**Aditya-L1**).

- **PSLV**: four stages (**solid and liquid alternately**); India's **workhorse**; first success **1994**; launched **104 satellites in one flight (15 February 2017)**. **GSLV**: uses a **cryogenic upper stage** (liquid hydrogen and liquid oxygen at very low temperatures); India's **own cryogenic stage** flew in **2014**. **LVM3** (GSLV Mk III): heaviest, about **4 tonnes to GTO**; used for **Chandrayaan-2 and -3** and **Gaganyaan**. **SSLV**: small satellites up to **500 kg**, quick and cheap.

3. Satellite families

Family	Purpose
INSAT / GSAT	**Communication, TV, telemedicine, weather** (INSAT-3D series), disaster warnings.
IRS / EOS	**Remote sensing** for crops, water, forests, cities, disasters: **Resourcesat, Cartosat (maps), RISAT (radar, sees through clouds), Oceansat**.
NavIC (IRNSS)	India's **regional** navigation system (**7 satellites**; covers India and about **1,500 km** around); **GAGAN** improves GPS accuracy for **aircraft**.

4. Major missions

Mission	Facts
Chandrayaan-1 (2008)	PSLV; its instruments found evidence of **water molecules** on the Moon.
Mangalyaan (MOM)	Launched **5 November 2013** by PSLV; entered **Mars orbit on 24 September 2014**: India became the **first country to succeed on its first attempt** and the **first in Asia** to reach Mars orbit.

Chandrayaan-2 and -3	**Chandrayaan-2 (2019)**: the orbiter works, but the lander **Vikram crashed**; **Chandrayaan-3** landed near the **south pole** on **23 August 2023**, the **first** to land there; the **Pragyan** rover confirmed **sulphur**.
Aditya-L1, XPoSat	**Aditya-L1** (launched September 2023, at L1 in January 2024) studies the **Sun's corona and solar wind**; **XPoSat** (January 2024) studies **X-ray polarisation** from black holes and neutron stars.
Recent and future	**SpaDeX** achieved **docking** of two satellites (**January 2025**), making India the **fourth** country to do so; **NISAR** (NASA–ISRO radar satellite) launched in **July 2025**; **Shubhanshu Shukla** became the first Indian to visit the **International Space Station (Axiom-4, June 2025)**; **Gaganyaan** human spaceflight; **Bharatiya Antariksh Station** planned by **2035**.

5. Centres and private players

- **Satish Dhawan Space Centre (SDSC SHAR), Sriharikota**, Andhra Pradesh: India's **main launch port** (on the east coast, near the equator, launching over the sea). **VSSC** (Thiruvananthapuram, rockets), **URSC** (Bengaluru, satellites), **LPSC** (liquid engines), **SAC** (Ahmedabad, payloads), **NRSC** (**Hyderabad**, remote-sensing data); a **second spaceport** at **Kulasekarapattinam** (Tamil Nadu) for SSLV.
- **Private sector: IN-SPACe (2020)** authorises private firms; **Skyroot's Vikram-S (November 2022)** was India's **first private rocket**; **Agnikul's Agnibaan (May 2024)** flew a **3D-printed engine**; **NSIL** sells ISRO's launch and satellite services.

EXAM TRAP

INCOSPAR 1962; **Thumba 21 Nov 1963**; **ISRO 15 Aug 1969**; **DoS 1972**; **Aryabhata 19 Apr 1975**; **SLV-3 1980** (Rohini, Kalam); **PSLV** (1994; 104 sats 2017); **GSLV** cryogenic (own 2014); **LVM3**; **SSLV**; orbits **SSPO 600–800 km**, **GEO 35,786 km**, **L1 1.5 million km**; **INSAT/GSAT, IRS, NavIC (7), GAGAN**; **Chandrayaan-1 2008** (water), **MOM 2013/2014**, **Chandrayaan-3 23 Aug 2023**, **Aditya-L1**, **XPoSat**, **SpaDeX 2025**, **NISAR 2025**, **Shukla 2025**; **Sriharikota (AP)**, **NRSC Hyderabad**; **Vikram-S 2022**, **Agnibaan 2024**.

Test yourself: 5 questions (tap an option to see the answer)

PRACTICE

Consider the following statements:
1. The PSLV is mainly used to place remote-sensing satellites in sun-synchronous polar orbits.
2. The GSLV uses a cryogenic upper stage.
3. Aryabhata, India's first satellite, was launched by a Soviet rocket in 1975.
4. NavIC is a global navigation system covering the whole world like GPS.
Which of the above are correct?

A. 1 and 2 only

B. 1, 2 and 3 only

C. 2, 3 and 4 only

D. 1, 2, 3 and 4

Answer: B. NavIC is a regional system covering India and about 1,500 km around it.

PRACTICE

Match List-I (mission) with List-II (achievement or aim):
List-I: A. Mangalyaan B. Chandrayaan-1 C. Aditya-L1 D. XPoSat
List-II: 1. Study of the Sun from L1 2. Reached Mars orbit on the first attempt 3. X-ray polarimetry 4. Evidence of water molecules on the Moon

A. A-2, B-4, C-1, D-3

B. A-4, B-2, C-1, D-3

C. A-2, B-1, C-4, D-3

D. A-2, B-4, C-3, D-1

Answer: A. MOM: Mars first attempt; Chandrayaan-1: water; Aditya-L1: Sun; XPoSat: X-ray polarimetry.

PRACTICE

Which one of the following statements is NOT correct?

A. The National Remote Sensing Centre is located in Hyderabad.

B. The Satish Dhawan Space Centre is at Sriharikota in Andhra Pradesh.

C. Vikram-S was India's first privately built rocket to be launched.

D. A geostationary satellite orbits at about 600 km above the Earth.

Answer: D. A geostationary orbit is about 35,786 km above the equator; 600–800 km is the sun-synchronous range.

PRACTICE

Assertion (A): Communication satellites are usually placed in geostationary orbit.
Reason (R): A satellite in that orbit goes round once in 24 hours with the Earth and appears fixed over one place, so ground antennas need not track it.
Choose the correct answer:

A. Both A and R are true and R is the correct explanation of A

B. Both A and R are true but R is not the correct explanation of A

C. A is true but R is false

D. A is false but R is true

Answer: A. Appearing fixed is exactly why geostationary orbit suits communication.

PRACTICE

ISRO needs to launch a 1,500-kg remote-sensing satellite into a sun-synchronous polar orbit reliably and at low cost. Which launch vehicle is the best choice?

A. SSLV

B. PSLV

C. LVM3

D. A sounding rocket

Answer: B. PSLV is designed for such orbits; SSLV carries only up to 500 kg, and LVM3 is meant for heavy GTO payloads.

RECALL BOX: CLOSE THE NOTES AND WRITE THESE IN YOUR OWN WORDS

1. Write the history (1962, 1963, 1969, 1972, 1975, 1980). 2. Write the five launch vehicles and four orbits. 3. Write satellite families, missions with one result each, centres and private rockets.

NOTEBOOK PAGE · Day 19: write this from memory

Close the notes. Write these headings, fill every blank, then correct in a different colour.

HISTORY: Congress, Moderates and Extremists

Area	Write
Why nationalism rose	Seven causes, with Lytton's measures and the Ilbert Bill
Early associations	Seven bodies with year, place, founder
Congress founded	Date, place, delegates, president, Hume, Dufferin, safety-valve theory; early sessions and presidents (1885–1907), P. Ananda Charlu 1891

Moderates vs Extremists	Leaders, goals, methods, base, achievements, limits; causes of extremism; Calcutta 1906, Surat 1907

POLITY: Lokpal and Lokayukta

- Origin (Sweden 1809, Singhvi 1963, ARC 1966, 2011 movement, 2013 Act); composition, selection, term, removal; jurisdiction and PM limits; powers; Lokayukta (Sec 63, Maharashtra 1971, AP 1983); criticism.

SCIENCE: Space technology

Write: INCOSPAR 1962, ISRO 1969, Thumba 1963, Aryabhata 1975; five launch vehicles; four orbits; satellite families (INSAT/GSAT, IRS, NavIC); missions with year and one result; centres; private rockets.